

David Ming

978-501-3858 | dhming@umass.edu | davidming1161@gmail.com

EDUCATION

University Of Massachusetts

Bachelor of Science in Mechanical Engineering

Amherst, MA

2024 – 2028

SKILLS

Programming Languages: Python

Tools: SolidWorks, OnShape, MATLAB, PrusaSlicer, Moldex3D, Ansys

EXPERIENCE

BSC Technology

December 2024 – January 2025

Mechanical Engineering Intern

Shenzhen, China

- Developed 3D models and contributed to the prototyping process, translating design concepts into tangible product iterations
- Hands on factory floor experience in an injection molding and assembly environment, observing the full production lifecycle
- Analyzed injection molding flow simulations to evaluate manufacturability and provided design recommendations to improve production outcomes
- Reviewed BOMs and prepared DFM documentation to communicate design feedback to clients

PROJECTS

FPV Drone

2023

- Assembled a fully functional FPV drone from components including frame, flight controller, motors, and propellers
- Wired and soldered electrical components, ensuring clean and reliable connections
- Designed and 3D printed custom parts to ensure proper fit and integration between components

Robotic Arm

2026

- Designing a fully custom multi-joint robotic arm from scratch using SolidWorks
- Developing control software to manage joint movement and motion sequencing
- Integrating a camera module with computer vision capabilities for object detection and interaction

Cycloidal Drive

2026 – Present

- Designing a custom cycloidal drive a 20:1 reduction ratio for high torque, low backlash motion
- Modeled the full assembly in SolidWorks, including the eccentric disc geometry
- Selected off the shelf ball bearings for the ring gear rollers and output supports, sized to the drive's load and tolerance requirements